



US 20250286156A1

(19) **United States**

(12) **Patent Application Publication**
Mun

(10) **Pub. No.: US 2025/0286156 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **BATTERY PACK INCLUDING BMS
COMMUNICATING WIRELESSLY**

Publication Classification

(51) **Int. Cl.**

H01M 10/42 (2006.01)

H01Q 15/16 (2006.01)

(52) **U.S. Cl.**

CPC **H01M 10/425** (2013.01); **H01Q 15/16**
(2013.01); **H01M 2010/4271** (2013.01); **H01M**
2010/4278 (2013.01)

(71) Applicant: **Samsung SDI Co., Ltd.**, Yongin-si
(KR)

(72) Inventor: **Seongjin Mun**, Yongin-si (KR)

(73) Assignee: **Samsung SDI Co., Ltd.**, Yongin-si
(KR)

(21) Appl. No.: **18/769,181**

(22) Filed: **Jul. 10, 2024**

(30) **Foreign Application Priority Data**

Mar. 5, 2024 (KR) 10-2024-0031529

(57)

ABSTRACT

A battery pack includes a battery module, a module battery management system (BMS) configured to monitor the battery module, a main BMS configured to exchange radio signals with the module BMS through wireless communication, and a signal reflection unit configured to reflect the radio signals exchanged between the module BMS and the main BMS toward the module BMS or the main BMS.

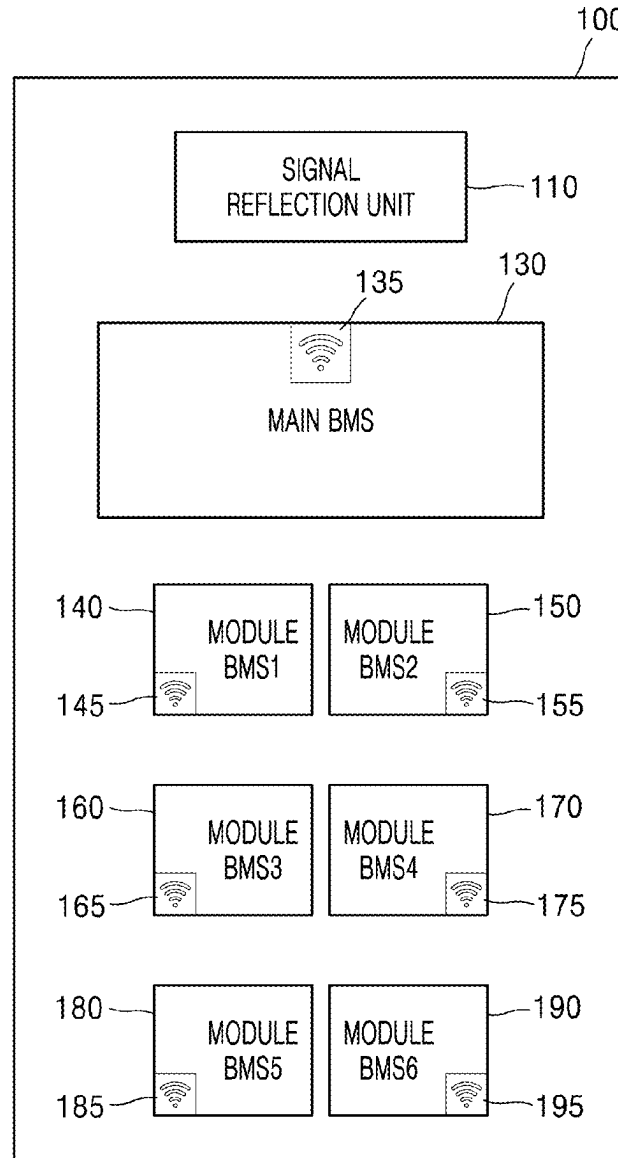


FIG. 1

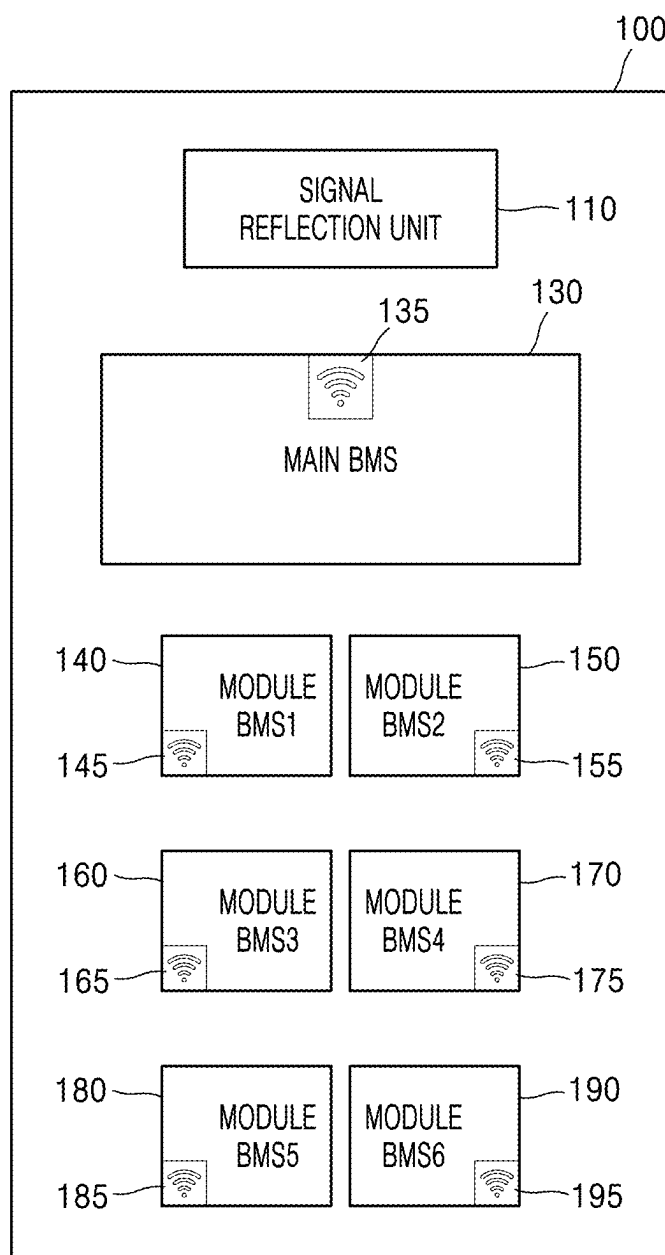


FIG. 2

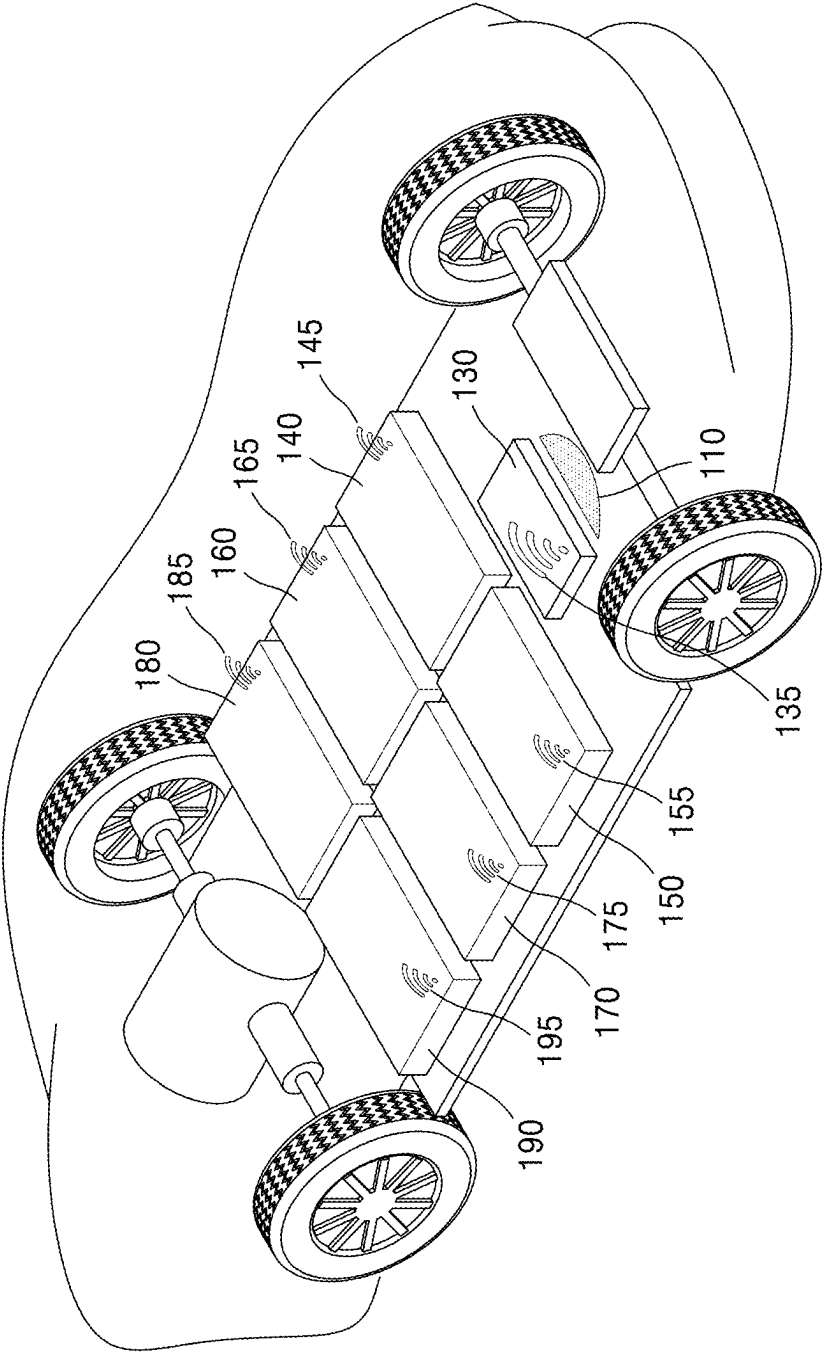


FIG. 3

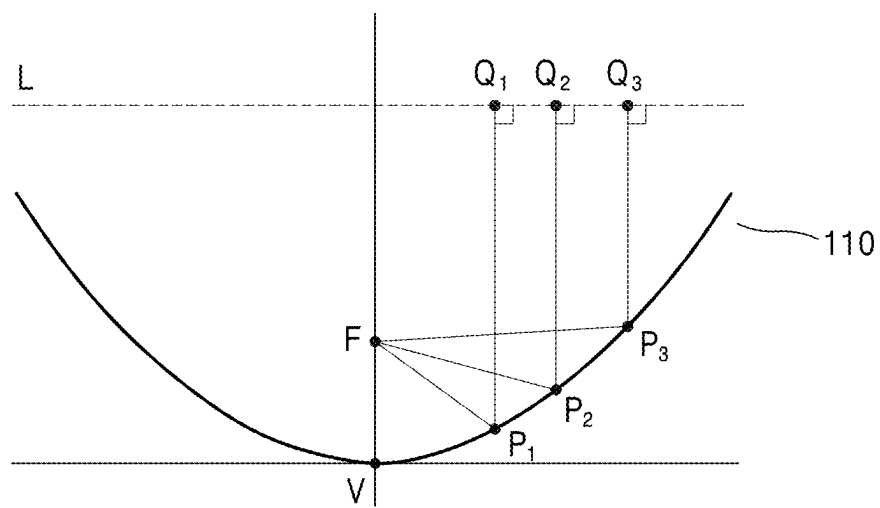
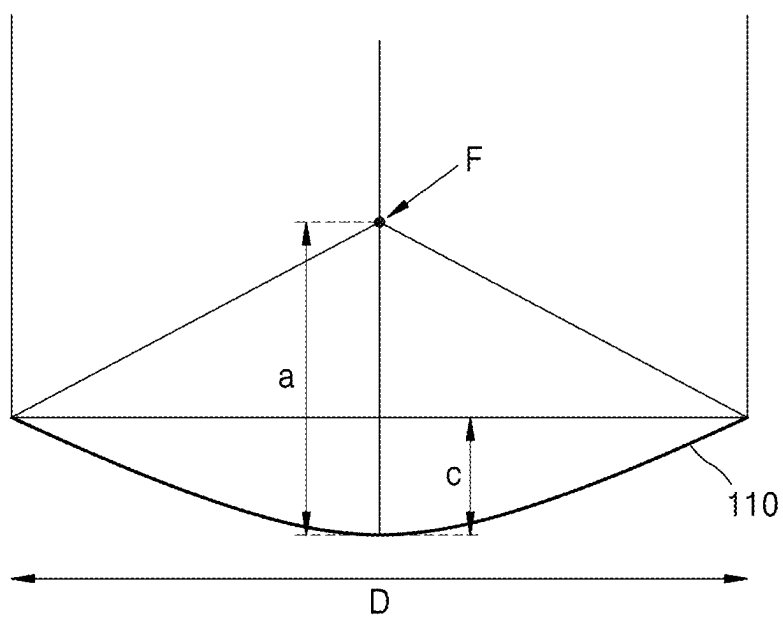


FIG. 4



**BATTERY PACK INCLUDING BMS
COMMUNICATING WIRELESSLY****CROSS-REFERENCE TO RELATED
APPLICATION**

[0001] The present application claims priority and the benefit under 35 U.S.C. § 119(a) of Korean Patent Application No. 10-2024-0031529, filed on Mar. 5, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference in its entirety.

FIELD

[0002] One or more embodiments relate to a battery pack including a battery management system (BMS) that communicates wirelessly.

BACKGROUND

[0003] A secondary battery can be charged and discharged, unlike a primary battery that cannot be recharged. Low-capacity secondary batteries are used in small portable electronic devices such as smartphones, feature phones, laptop computers, digital cameras, and camcorders, and high-capacity secondary batteries are used as motor-driving power sources, power-storing batteries, etc., for hybrid vehicles, electric vehicles, etc. Such a secondary battery may include an electrode assembly including a cathode and an anode, a case accommodating the electrode assembly, an electrode terminal connected to the electrode assembly, etc.

[0004] To configure battery packs for electric vehicles including such secondary batteries, wireless battery management system (BMS) technology is attracting attention. The wireless BMS has the advantage of not requiring cables for communication between modules. However, as wired communication is replaced with wireless communication, communication quality is inevitably affected by antenna sensitivity in reception or transmission situations, and if communication quality deteriorates, stable BMS operation becomes difficult to achieve.

[0005] The above-described information disclosed in the technology section that serves as the background of the present disclosure is only for improving the understanding of the background of the present disclosure, and thus may include information that does not constitute prior art.

SUMMARY

[0006] One or more embodiments include a battery pack including a battery management system (BMS) that communicates wirelessly.

[0007] However, technical problems to be solved by the present disclosure are not limited to the above-mentioned problem, and other problems not mentioned may be clearly understood by those of ordinary skill in the art from the description of the present disclosure described below.

[0008] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the present disclosure.

[0009] According to one or more embodiments, a battery pack includes a battery module, a module battery management system (BMS) configured to monitor the battery module, a main BMS configured to exchange radio signals with the module BMS through wireless communication, and a signal reflection unit configured to reflect the radio signals

exchanged between the module BMS and the main BMS toward the module BMS or the main BMS.

[0010] According to an example, the main BMS may include a main BMS antenna unit configured to exchange radio signals with the module BMS.

[0011] According to another example, the signal reflection unit may include a structure having a parabolic curved surface.

[0012] According to another example, the signal reflection unit may be located at a distance from the main BMS antenna unit of the main BMS, the distance being equal to a focal length of the parabolic curved surface such that the main BMS antenna unit is located at a focus of the parabolic curved surface.

[0013] According to another example, the signal reflection unit may include a structure configured to reflect, toward the module BMS antenna unit, radio signals transmitted from the main BMS antenna unit.

[0014] According to another example, the signal reflection unit may include a structure configured to reflect, toward the main BMS antenna unit, radio signals transmitted from the module BMS antenna unit.

[0015] According to another example, the signal reflection unit may include a structure having a surface facing the main BMS antenna unit, the surface being plated with conductive metal.

[0016] According to another example, the signal reflection unit may include a structure in which the focal length of the parabolic curved surface is preset based on positions of the main BMS antenna unit and the module BMS antenna unit.

[0017] According to another example, the signal reflection unit may include a structure in which the focal length of the parabolic curved surface is preset such that the radio signals transmitted from the main BMS antenna unit are reflected and oriented toward the module BMS antenna unit.

[0018] According to one or more embodiments, a method of operating a battery pack including a battery module, a module battery management system (BMS), a main BMS, and a signal reflection unit includes monitoring, by the module BMS, the battery module, exchanging, by the main BMS, radio signals with the module BMS through wireless communication, and reflecting, by the signal reflection unit, the radio signals exchanged between the module BMS and the main BMS toward the module BMS or the main BMS.

[0019] According to an example, the module BMS may include a module BMS antenna unit, and the exchanging may include the module BMS antenna unit exchanging radio signals with the main BMS.

[0020] According to another example, the main BMS may include a main BMS antenna unit, and the exchanging may further include the main BMS antenna unit exchanging radio signals with the module BMS.

[0021] According to another example, the signal reflection unit may include a structure having a parabolic curved surface.

[0022] According to another example, the signal reflection unit may be located at a distance from the main BMS antenna unit of the main BMS, the distance being equal to a focal length of the parabolic curved surface such that the main BMS antenna unit is located at a focus of the parabolic curved surface.

[0023] According to another example, the reflecting may include reflecting, by the signal reflection unit, toward the module BMS antenna unit, radio signals transmitted from the main BMS antenna unit.

[0024] According to another example, the reflecting may include reflecting, by the signal reflection unit, toward the main BMS antenna unit, radio signals transmitted from the module BMS antenna unit.

[0025] According to another example, the signal reflection unit may include a structure having a surface facing the main BMS antenna unit, the surface being plated with conductive metal.

[0026] According to another example, the signal reflection unit may include a structure in which the focal length of the parabolic curved surface is preset based on positions of the main BMS antenna unit and the module BMS antenna unit.

[0027] According to another example, the signal reflection unit may include a structure in which the focal length of the parabolic curved surface is preset such that the reflecting includes the radio signals transmitted from the main BMS antenna unit being reflected and oriented towards the module BMS antenna unit.

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings.

[0029] The following drawings attached to this specification illustrate preferred embodiments of the present disclosure, and serve to further understand the technical idea of the present disclosure together with the detailed description of the present disclosure described later, so the present disclosure should not be construed as being limited to the matters shown in such drawings.

[0030] FIG. 1 schematically shows a battery pack according to embodiments of the present disclosure;

[0031] FIG. 2 schematically shows a vehicle body including a battery pack according to embodiments of the present disclosure;

[0032] FIG. 3 is a diagram illustrating a structure using which a radio signal is reflected from a signal reflection unit, according to embodiments of the present disclosure; and

[0033] FIG. 4 is a diagram illustrating a focal length of a parabolic curved surface of a signal reflection unit, according to embodiments of the present disclosure.

DETAILED DESCRIPTION

[0034] Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects of the present description. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” if preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

[0035] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompa-

nying drawings. The terms and words used in the present specification and claims described above should not be construed as being limited to ordinary or dictionary meanings, and should be interpreted as meanings and concepts consistent with the technical idea of the present disclosure based on the principle that the present inventors may appropriately define the concept of the terms to describe their invention in the best way. Therefore, it should be understood that the configurations shown in the drawings and embodiments described in this specification are merely the most preferred embodiments of the present disclosure, and do not represent all of the technical ideas of the present disclosure, such that there may be various equivalents and variations that replace them at the time of filing the present application. If used herein, “comprise, include” and/or “comprising, including” specify mentioned shapes, numbers, steps, operations, members, components, and/or presence of these groups, and do not exclude the presence or addition of one or more different shapes, numbers, operations, members, components, and/or groups. If embodiments of the present disclosure are described, “can” or “may” may include “one or more embodiments of the present disclosure”.

[0036] To help understanding of the present disclosure, the accompanying drawings are not shown according to the actual scale, but the dimensions of some components may be exaggerated. The same reference numeral may be given to the same component in different embodiments.

[0037] The statement that two comparison targets are ‘the same’ as each other may mean that they are ‘substantially the same’ as each other. Thus, a case where they are ‘substantially the same’ as each other may include a case where they have a deviation regarded as a low level, e.g., a deviation of 5% or less. If a uniform parameter is uniform in a predetermined area, it may mean that it is uniform from an average point of view.

[0038] Although first, second, etc., may be used to describe various components, these components are not limited by these terms. These terms are only used to distinguish one component from other components, and unless specifically stated to the contrary, a first component may be a second component.

[0039] Throughout the specification, unless specially stated to the contrary, each component may be singular or plural.

[0040] If a component is arranged on “a top portion (or a bottom portion)” of another component or “on (or under)” the other component, it may mean not only a case where the component is arranged adjacent to a top surface (or a bottom surface) of the other component, but also a case where another component may be interposed between the other component and the component arranged on (or under) the other component.

[0041] If a component is described as being “connected”, “coupled”, or “connected” to another component, it should be understood that the components are directly connected or connectable to each other, but another component may be “interposed” between the components, or the components may be may be “connected”, “coupled”, or “connected” to each other through another component. If a portion is electrically coupled to another portion, this may include not only a case where they are directly connected to each other, but also a case where they are connected with another element therebetween.

[0042] Throughout the specification, “A and/or B” may mean A, B, or A and B unless specially stated otherwise. That is, “and/or” may include all or any combination of a plurality of items listed. “C to D” may mean at least C but not more than D, unless specially stated otherwise.

[0043] FIG. 1 schematically shows a battery pack according to embodiments of the present disclosure.

[0044] Referring to FIG. 1, a battery pack 100 according to embodiments of the present disclosure may include a signal reflection unit 110, a main battery management system (BMS) 130, module BMSs 140, 150, 160, 170, 180, and 190, a main BMS antenna unit 135, and module BMS antenna units 145, 155, 165, 175, 185, and 195.

[0045] The battery pack 100 according to embodiments of the present disclosure may include at least one battery module and a pack housing in which an accommodation space is formed to accommodate the at least one battery module. The battery module may include a plurality of battery cells and a module housing. The battery cells may be accommodated in a stacked form in the module housing. Each battery cell may include an anode lead and a cathode lead. The battery cell may be of a circular type, a prismatic type, or a pouch type, depending on a shape of a battery.

[0046] The battery module may include at least one battery cell that may be a chargeable secondary battery. For example, the battery cell may include at least one selected from a group including a nickel-cadmium battery, a lead acid battery, a nickel metal hydride battery (NiMH), a lithium ion battery, a lithium polymer battery, etc.

[0047] The number of battery cells included in the battery module and a connection scheme thereof may be determined based on a power amount, a voltage, etc., required for the battery pack 100. For example, the battery cells may be connected in parallel to each other or in parallel and in series to each other. The battery pack 100 may include a plurality of battery modules connected in series, in parallel, or in series and in parallel. The battery module may include one battery cell.

[0048] The battery pack 100 according to embodiments of the present disclosure may include a main BMS 130 and module BMSs 140, 150, 160, 170, 180, and 190. The main BMS 130 and the module BMSs 140, 150, 160, 170, 180, and 190 may indicate BMSs for managing a battery pack and a battery module. For example, the BMS may include a detection device, a balancing device, and a control device. For example, referring to FIG. 1, the module BMSs 140, 150, 160, 170, 180, and 190 may respectively monitor battery modules. For example, the first module BMS 140 may monitor a first battery module, the second BMS 150 may monitor a second battery module, the third module BMS 160 may monitor a third battery module, the fourth module BMS 170 may monitor a fourth battery module, the fifth module BMS 180 may monitor a fifth battery module, and the sixth module BMS 190 may monitor a sixth battery module. The main BMS 130 may manage the battery pack 100. The main BMS 130 and the module BMSs 140, 150, 160, 170, 180, and 190 may exchange radio signals with each other through wireless communication.

[0049] The detection device may sense a state (a voltage, a current, a temperature, etc.) of a battery to detect state information indicating the state of the battery. The detection device may detect a voltage of each cell or each battery module of the battery. The detection device may detect current flowing through the battery module or each battery

module of the battery pack. The detection device may detect a temperature of a cell and/or module or an ambient temperature in at least a point of the battery.

[0050] The balancing device may perform a balancing operation on a battery module and/or cells of the battery.

[0051] The control device may receive state information (a voltage, a current, a temperature, etc.) of a battery module from the detection device. The control device may monitor and calculate a state (a voltage, a current, a temperature, a state of charge (SOC), a lifetime, a state of health (SOH), etc.) of a battery module, based on the state information received from the detection device. The control device may perform a control function (e.g., temperature control, balancing control, charging/discharging control, etc.), a protection function (e.g., prevention of over-discharge, over-charge, and over-current, short-circuit, fire extinguishment, etc.), and so forth, based on a state monitoring result. The control device may also perform a wired or wireless communication function with an external device (e.g., a higher-level controller, a vehicle, a charger, a PCS, etc.) of a battery pack. In addition, the control device may control a charging/discharging operation and a protection operation of the battery. To this end, the control device may include a charging/discharging controller, a balancing controller, and a protection unit.

[0052] The BMS, which is a system that monitors a state of a battery and performs diagnosis and control, communication, and protection, may calculate a charging/discharging state, calculate a lifetime or a state of health (SOH) of the battery, block power of the battery (relay control) if required, control thermal management (cooling, heating, etc.), perform a high-voltage interlocking function, and detect or calculate an insulation and short-circuit state.

[0053] The main BMS 130 according to embodiments of the present disclosure may include a main BMS antenna unit 135. The module BMSs 140, 150, 160, 170, 180, and 190 may respectively include module BMS antenna units 145, 155, 165, 175, 185, and 195. For example, the main BMS antenna unit 135 and the module BMS antenna units 145, 155, 165, 175, 185, and 195 may indicate antenna devices that perform wireless communication. For example, the main BMS 130 and the module BMSs 140, 150, 160, 170, 180, and 190 may exchange radio signals with each other through wireless communication by using the main BMS antenna unit 135 and the module BMS antenna units 145, 155, 165, 175, 185, and 195. For example, the antenna device and a scheme of the wireless communication used may be a known device and scheme, respectively.

[0054] The battery pack 100 according to embodiments of the present disclosure may include the signal reflection unit 110. For example, the signal reflection unit 110 may be a structure having a surface facing an antenna that radiates a radio signal, in which the surface is plated with conductive metal to reflect the radio signal. For example, the signal reflection unit 110 may indicate a structure that reflects, toward the module BMSs 140, 150, 160, 170, 180, and 190 or the main BMS 130, radio signals exchanged between the module BMSs 140, 150, 160, 170, 180, and 190 and the main BMS 130.

[0055] FIG. 2 schematically shows a vehicle body including a battery pack according to embodiments of the present disclosure.

[0056] Referring to FIG. 2, a structure is shown in which the battery pack 100 according to embodiments of the

present disclosure is provided under a vehicle body. For example, the battery pack 100 may include six battery modules. For example, the six battery modules may be electrically connected to each other in series/parallel or in a serial/parallel combination manner. As shown in FIG. 2, the module BMSs 140, 150, 160, 170, 180, and 190 may be provided in respective battery modules. The module BMSs 140, 150, 160, 170, 180, and 190 may respectively include the module BMS antenna units 145, 155, 165, 175, 185, and 195 that exchange radio signals with the main BMS 130. The main BMS 130 may include the main BMS antenna unit 135 that exchange radio signals with the module BMS antenna units 145, 155, 165, 175, 185, and 195 of the respective module BMSs 140, 150, 160, 170, 180, and 190.

[0057] The signal reflection unit 110 may be a structure having a parabolic curved surface. For example, the signal reflection unit 110 may be a structure in which a surface facing the main BMS antenna unit 135 or the module antenna units 145, 155, 165, 175, 185, and 195 has a parabolic curved surface that is plated with conductive metal.

[0058] FIG. 3 is a diagram illustrating a structure using which a radio signal is reflected from a signal reflection unit, according to embodiments of the present disclosure. FIG. 4 is a diagram illustrating a focal length of a parabolic curved surface of a signal reflection unit, according to embodiments of the present disclosure.

[0059] Referring to FIGS. 2 to 4 together, the signal reflection unit 110 according to embodiments of the present disclosure may have a structure that uses a parabolic curved surface structure of a parabolic antenna, such as a satellite antenna, thereby improving sensitivity by enhancing the directivity of a radio signal in radio signal reception or transmission of the main BMS antenna unit 135 of the main BMS 130.

[0060] For example, as shown in FIGS. 3 and 4, the signal reflection unit 110 may be disposed such that the main BMS antenna unit 135 is located at a focus F of the signal reflection unit 110. For example, the signal reflection unit 110 may be located at a distance equal to a focal length a of a parabolic curved surface from the main BMS antenna unit 135 of the main BMS 130 such that the main BMS antenna unit 135 is located at the focus F of the parabolic curved surface. For example, a curvature of the parabolic curved surface of the signal reflection unit 110 may be calculated in advance and preset.

[0061] The signal reflection unit 110 may have a structure that reflects, toward the module BMS antenna units 145, 155, 165, 175, 185, and 195, radio signals transmitted from the main BMS antenna unit 135. The signal reflection unit 110 may have a structure that reflects, toward the main BMS antenna unit 135, radio signals transmitted from the module BMS antenna units 145, 155, 165, 175, 185, and 195. For example, as shown in FIGS. 2 and 3, a radio signal transmitted from the main BMS antenna unit 135 located at the focus F may be reflected from reflection points P_1 , P_2 , and P_3 of the signal reflection unit 110 and may be propagated toward points Q_1 , Q_2 , and Q_3 of the module BMS antenna units 145, 155, 165, 175, 185, and 195. For example, the module BMS antenna units 145, 155, 165, 175, 185, and 195 may be disposed at the points Q_1 , Q_2 , and Q_3 toward which the radio signal is oriented after being reflected from the reflection points P_1 , P_2 , and P_3 .

[0062] The signal reflection unit 110 may be a structure in which a surface facing the main BMS antenna unit 135 is plated with conductive metal. In the signal reflection unit 110 according to some embodiments of the present disclosure, an inner surface in the semi-circular shape is treated with conductive metal plating to serve as a radio signal reflection plate, thereby improving directivity of a radio signal toward the main BMS antenna unit 135 or the module BMS antenna units 145, 155, 165, 175, 185, and 195 and thus enhancing the sensitivity of reception or transmission of a radio signal.

[0063] The signal reflection unit 110 may be a structure in which a focal length of a parabolic curved surface is preset based on the positions of the main BMS antenna unit 135 and the module BMS antenna units 145, 155, 165, 175, 185, and 195.

[0064] The signal reflection unit 110 may be a structure in which the focal length of the parabolic curved surface is preset such that the radio signals transmitted from the main BMS antenna unit 135 are reflected and oriented toward the module BMS antenna units 145, 155, 165, 175, 185, and 195.

[0065] For example, as shown in FIG. 4, the main BMS antenna unit 135 may be disposed at the focus F of the parabolic curved surface of the signal reflection unit 110 according to embodiments of the present disclosure.

[0066] The curvature of the parabolic curved surface of the signal reflection unit 110 may be calculated as below.

$$a=(D*D)/(16*c)$$

[0067] Herein, a may be a focal length of the signal reflection unit 110, c may be a depth of the signal reflection unit 110, and D may be a width of the signal reflection unit 110.

[0068] For example, in the signal reflection unit 110 having D of about 10 cm and c of about 1 cm, the position of the main BMS antenna unit 135 may be located at a focal length of about 6.25 cm from the center of the signal reflection unit 110.

[0069] According to the present disclosure, the sensitivity of the antenna in charge of reception or transmission of data in a wireless BMS may be improved through the signal reflection unit included in the wireless BMS.

[0070] The signal reflection unit according to the present disclosure may be a structure in which a surface visible to the antenna and having a semi-circular shape is plated with metal to improve directivity of a signal in a desired direction in wireless reception or transmission via the antenna, thereby enhancing wireless sensitivity, and the principle of a parabolic antenna having a parabolic curved surface is applied to a wireless BMS, thereby improving the performance of a wireless antenna.

[0071] While the present disclosure describes some embodiments and drawings, the present disclosure is not limited thereby and various modifications and changes may be made by those of ordinary skill in the art within the technical spirit of the present disclosure and the equivalent range to the claims set forth below.

[0072] Various embodiments of the present disclosure may be implemented in the form of a computer program executable on a computer through various components, and the computer program may be recorded on a computer-readable medium. The medium may continuously store an executable program or temporarily store the same for execu-

tion or downloading. The medium may include various recording means or storage means in a form of single hardware or a combination of several hardware, and may be distributed over a network without being limited to a medium directly connected to a certain computer system. Examples of the medium may include a magnetic medium such as a hard disk, a floppy disk, and a magnetic tape, an optical recording medium such as a compact disc (CD)-read-only memory (ROM) and a digital versatile disk (DVD), a magneto-optical medium such as a floptical disk, ROM, random-access memory (RAM), flash memory, etc., to store a program instruction. Other examples of the medium may include a recording medium or a storage medium managed by an app store that distributes applications, a site that supplies or distributes various software, a server, etc.

[0073] In the specification, the term “unit”, “module”, etc., may be a hardware component like a processor or a circuit, and/or a software component executed by a hardware component like a processor. For example, “unit”, “module”, etc., may be implemented by components such as software components, object-oriented software components, class components, and task components, processes, functions, properties, procedures, subroutines, segments of program code, drivers, firmware, microcode, circuits, data, a database, data structures, tables, arrays, and variables.

[0074] According to the present disclosure, in a battery pack including a BMS that communicates wirelessly, the reception or sensitivity of an antenna included in the BMS may be efficiently improved.

[0075] However, effects that may be obtained through the present disclosure are not limited to the above-described effects, and other technical effects not mentioned may be clearly understood by those of ordinary skill in the art from the description of the present disclosure described below.

[0076] It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure as defined by the following claims.

[0077] What is claimed is:

1. A battery pack comprising:
 - a battery module;
 - a module battery management system (BMS) configured to monitor the battery module;
 - a main BMS configured to exchange radio signals with the module BMS through wireless communication; and
 - a signal reflection unit configured to reflect the radio signals exchanged between the module BMS and the main BMS toward the module BMS or the main BMS.
2. The battery pack as claimed in claim 1, wherein the module BMS comprises a module BMS antenna unit configured to exchange radio signals with the main BMS.
3. The battery pack as claimed in claim 2, wherein the main BMS comprises a main BMS antenna unit configured to exchange radio signals with the module BMS.

4. The battery pack as claimed in claim 3, wherein the signal reflection unit comprises a structure having a parabolic curved surface.

5. The battery pack as claimed in claim 4, wherein the signal reflection unit is located at a distance from the main BMS antenna unit of the main BMS, the distance being equal to a focal length of the parabolic curved surface such that the main BMS antenna unit is located at a focus of the parabolic curved surface.

6. The battery pack as claimed in claim 5, wherein the signal reflection unit comprises a structure configured to reflect, toward the module BMS antenna unit, radio signals transmitted from the main BMS antenna unit.

7. The battery pack as claimed in claim 5, wherein the signal reflection unit comprises a structure configured to reflect, toward the main BMS antenna unit, radio signals transmitted from the module BMS antenna unit.

8. The battery pack as claimed in claim 5, wherein the signal reflection unit comprises a structure having a surface facing the main BMS antenna unit, the surface being plated with conductive metal.

9. The battery pack as claimed in claim 5, wherein the signal reflection unit comprises a structure in which the focal length of the parabolic curved surface is preset based on positions of the main BMS antenna unit and the module BMS antenna unit.

10. The battery pack as claimed in claim 9, wherein the signal reflection unit comprises a structure in which the focal length of the parabolic curved surface is preset such that the radio signals transmitted from the main BMS antenna unit are reflected and oriented toward the module BMS antenna unit.

11. A method of operating a battery pack comprising a battery module, a module battery management system (BMS), a main BMS, and a signal reflection unit, the method comprising:

- monitoring, by the module BMS, the battery module;
- exchanging, by the main BMS, radio signals with the module BMS through wireless communication; and
- reflecting, by the signal reflection unit, the radio signals exchanged between the module BMS and the main BMS toward the module BMS or the main BMS.

12. The method as claimed in claim 11, wherein the module BMS comprises a module BMS antenna unit, and the exchanging comprises the module BMS antenna unit exchanging radio signals with the main BMS.

13. The method as claimed in claim 12, wherein the main BMS comprises a main BMS antenna unit, and the exchanging further comprises the main BMS antenna unit exchanging radio signals with the module BMS.

14. The method as claimed in claim 13, wherein the signal reflection unit comprises a structure having a parabolic curved surface.

15. The method as claimed in claim 14, wherein the signal reflection unit is located at a distance from the main BMS antenna unit of the main BMS, the distance being equal to a focal length of the parabolic curved surface such that the main BMS antenna unit is located at a focus of the parabolic curved surface.

16. The method as claimed in claim 15, wherein the reflecting comprises reflecting, by the signal reflection unit, toward the module BMS antenna unit, radio signals transmitted from the main BMS antenna unit.

17. The method as claimed in claim **15**, wherein the reflecting comprises reflecting, by the signal reflection unit, toward the main BMS antenna unit, radio signals transmitted from the module BMS antenna unit.

18. The method as claimed in claim **15**, wherein the signal reflection unit comprises a structure having a surface facing the main BMS antenna unit, the surface being plated with conductive metal.

19. The method as claimed in claim **15**, wherein the signal reflection unit comprises a structure in which the focal length of the parabolic curved surface is preset based on positions of the main BMS antenna unit and the module BMS antenna unit.

20. The method as claimed in claim **19**, wherein the signal reflection unit comprises a structure in which the focal length of the parabolic curved surface is preset such that the reflecting comprises the radio signals transmitted from the main BMS antenna unit being reflected and oriented toward the module BMS antenna unit.

* * * * *